



EFFICACY AND SAFETY OF PUSH-DOSE NOREPINEPHRINE FOR INTRAOPERATIVE HYPOTENSION MANAGEMENT IN NON-OBSTETRIC SURGERY: A SYSTEMATIC REVIEW

R. Suseno Aji, Calcarina Fitriani Retno Wisudarti, Djayanti Sari

Department of Anesthesiology & Intensive Therapy · Faculty of Medicine, Public Health and Nursing
Universitas Gadjah Mada / RSUP Dr. Sardjito, Yogyakarta, Indonesia

BACKGROUND

Intraoperative hypotension (IOH), defined as MAP <65 mmHg, occurs in up to **30%** of non-cardiac surgeries and is independently associated with myocardial injury, acute kidney injury, and increased 30-day mortality. Push-dose (bolus) norepinephrine (NE) has emerged as a practical vasopressor; however, high-quality evidence on its efficacy and safety remains limited.

METHODS

Design: Systematic review · PRISMA 2020

Databases: PubMed, Scopus, ProQuest

PICO: Adults ≥18 yr · Non-obstetric surgery · RCT design

Quality: Cochrane RoB 2 tool

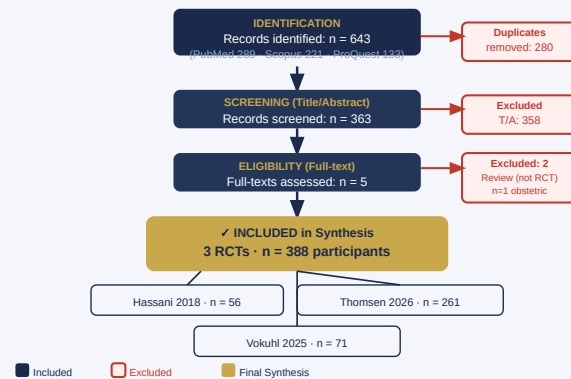
Certainty: GRADE framework

Search string: "Norepinephrine" AND ("push dose" OR "bolus") AND ("intraoperative hypotension" OR "surgery")

OBJECTIVE

To systematically evaluate the **efficacy and safety** of push-dose NE for IOH in non-obstetric surgery compared to other vasopressors, following **PRISMA 2020** guidelines.

PRISMA 2020 FLOW DIAGRAM



RESULTS – SUMMARY

3 RCTs · n = 388 · Push-dose NE **10 mcg/bolus** (10 mcg/mL, peripheral IV)

- vs. Ephedrine:** NE superior MAP (98.9 vs 91.5 mmHg; **p<0.001**), fewer hypotension episodes (**p=0.005**)
- vs. Continuous NE (low-mod risk):** Comparable AUC MAP <65 (p=0.070), 66% less drug dose (**p<0.001**)
- vs. Continuous NE (high risk):** Infusion better ARV-MAP stability (**p<0.001**, d=1.00)
- Safety:** No significant HTN overshoot or arrhythmia; peripheral IV safe — zero extravasation

MAP reversal - Low

HTN overshoot - Low

Time to target - Very Low

RISK OF BIAS 2 (COCHRANE) – TRAFFIC LIGHT

	D1	D2	D3	D4	D5	Overall
Hassani 2018 · n=56 NE vs Epi	⚠	✓	✓	✓	✓	⚠
Thomsen 2026 · n=261 INDUCT Trial	✓	⚠	✓	✓	✓	⚠
Vokuhl 2025 · n=71 High-risk surgery	✓	⚠	✓	✓	✓	⚠
	Random.	Deviations	Missing	Outcome	Selection	Overall

Summary of Included Studies

Study	Population	Intervention vs Comparator	Key Findings (MAP / Safety)
Hassani (2018) Iran · RCT · n=56	Hypertensive, GA Spinal Surgery	NE 10 mcg bolus vs Ephedrine 5 mg bolus	NE superior MAP (98.9 vs 91.5 mmHg; p<0.001), fewer vasopressor doses, less bradycardia
Thomsen (2026) Denmark · RCT · n=261	ASA ≥2, Low-Moderate Risk, Noncardiac	NE Bolus 10 mcg/mL vs Continuous Infusion	AUC MAP<65 comparable (p=0.070); 66% lower cumulative dose (p<0.001)
Vokuhl (2025) Germany · RCT · n=71	ASA II–IV, High-Risk Elective Major Surgery	NE Bolus 10 mcg/mL vs Continuous Infusion	Continuous infusion better haemodynamic stability (ARV-MAP; p<0.001 , d=1.00); peripheral IV safe

CONCLUSIONS

Push-dose NE **10 mcg** via peripheral IV is **effective and safe** for IOH in non-obstetric surgery. It is **superior to ephedrine** and offers comparable efficacy to continuous infusion in **low-to-moderate risk** patients. For **high-risk patients** with arterial monitoring, continuous infusion may provide better haemodynamic stability. **GRADE Low** (⊕⊕○○) — multicenter RCTs with standardised outcome definitions are needed.

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